

CSA1713 – Artificial Intelligence

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Slot – C

8. TRAVELLING SALESMAN PROBLEM

Code:

```
print("B. Deepthi Reddy")
print(192312288)
print("TRAVELLING SALESMAN PROBLEM")
from itertools import permutations

d = [[0,10,15,20],
      [10,0,35,25],
      [15,35,0,30],
      [20,25,30,0]]

cost = 999
path = ()

for p in permutations([1,2,3]):
    c = d[0][p[0]] + d[p[0]][p[1]] + d[p[1]][p[2]] + d[p[2]][0]
    if c < cost:
```

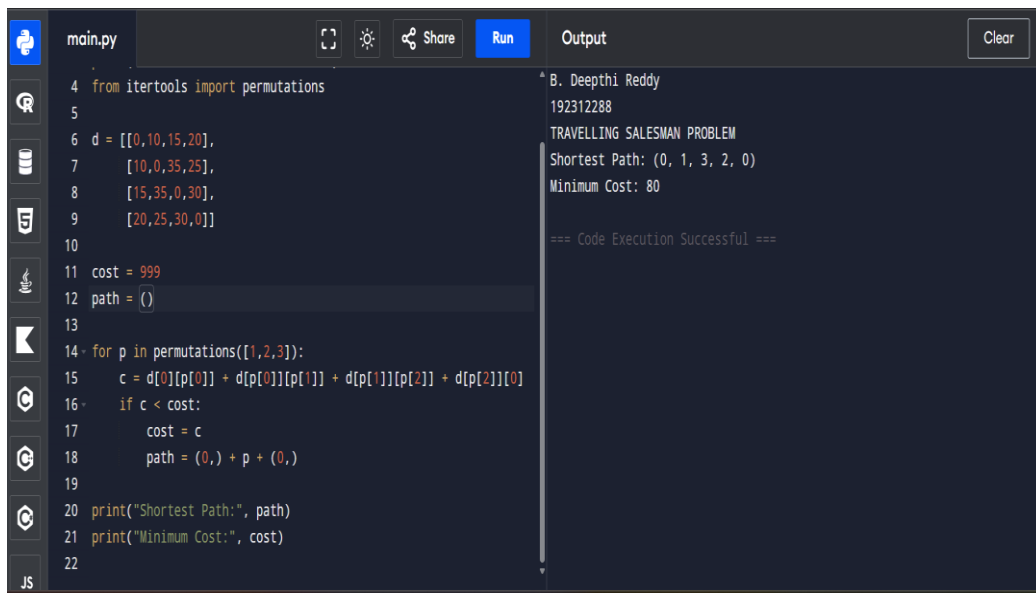
```
cost = c
```

```
path = (0,) + p + (0,)
```

```
print("Shortest Path:", path)
```

```
print("Minimum Cost:", cost)
```

Output:



The screenshot shows a code editor with a file named `main.py`. The code defines a distance matrix `d` and iterates through all permutations of nodes [1, 2, 3] to find the shortest path and minimum cost. The output window displays the user's name, ID, problem title, the shortest path, and the minimum cost, followed by a success message.

```
main.py
4 from itertools import permutations
5
6 d = [[0,10,15,20],
7       [10,0,35,25],
8       [15,35,0,30],
9       [20,25,30,0]]
10
11 cost = 999
12 path = ()
13
14 for p in permutations([1,2,3]):
15     c = d[0][p[0]] + d[p[0]][p[1]] + d[p[1]][p[2]] + d[p[2]][0]
16     if c < cost:
17         cost = c
18         path = (0,) + p + (0,)
19
20 print("Shortest Path:", path)
21 print("Minimum Cost:", cost)
22
```

Output:

```
B. Deepthi Reddy
192312288
TRAVELLING SALESMAN PROBLEM
Shortest Path: (0, 1, 3, 2, 0)
Minimum Cost: 80

=== Code Execution Successful ===
```